13 – A GUIDED EXAMPLE

True/False Questions

Mark the following statements as True (T) or False (F):

1. The automation system described in the chapter controls only ventilation and lighting.
2. The control panel allows users to configure the heating system's temperature settings.
3. A temperature sensor is responsible for measuring room occupancy.
4. The lighting subsystem reacts to occupancy changes by turning lights on or off.
5. The regulator in the automation system only processes temperature readings.

Multiple-Choice Questions

Choose the correct answer(s):

6. What are the major subsystems of the automation system described?
   a) Heating
   b) Security
   c) Ventilation
   d) Lighting

7. What factors determine the operation of the ventilation system?
   a) Air temperature
   b) Moisture level
   c) Occupancy in the room
   d) Pre-set time range
8. According to the automation rules, when should the radiator turn on?
   a) When the room temperature is below 18°C
   b) When the moisture level is above 70%
   c) When the user manually turns it on
   d) When the room temperature is above 22°C
9. What is the function of a regulator in the automation system?
   a) It replaces broken sensors
   b) It processes readings from sensors and applies automation rules

c) It switches all subsystems on and off manually

d) It only controls the heating system

Matching Exercise

Match the components with their correct function:

Component	Function
10. Radiator	a) Detects whether a room is occupied
11. Ventilator	b) Measures the room temperature
12. Moisture Sensor	c) Turns on/off based on temperature settings
13. Light Sensor	d) Controls airflow based on moisture levels
14. Occupancy Sensor	e) Measures the level of humidity in the air

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Comprehension Questions for Discussion

1. **Understanding the System**
   - What is the main objective of the home automation system described in the chapter?
   - How does the system ensure efficient temperature regulation in different rooms?
   - What role does the control panel play in managing the automation system?
2. **Problem Decomposition & Computational Thinking**
   - How does the chapter break down the automation problem into smaller components?
   - What are the key entities identified in the system, and why is it important to classify them?
   - What is the difference between **automation** and **control** in the context of this system?
3. **Rules and Decision-Making**
   - The automation system operates based on specific rules. Can you identify these rules and explain how they apply to different subsystems?
   - Why is it necessary for the **lighting system** to have a countdown feature before turning off the lights? Can you think of a real-world scenario where this feature would be beneficial?
   - How does the system handle **multiple conditions** at the same time (e.g., moisture level and scheduled time for ventilation)?
4. **Practical Applications and Limitations**
   - What are some possible real-world improvements that could make this system more effective?
   - Can you think of any drawbacks or challenges in implementing such a home automation system?
   - How would adding **machine learning** or **AI** improve the system's efficiency?

5. **Ethical & Security Considerations**
    - Home automation involves **sensors** collecting data. What potential privacy and security concerns might arise?
    - Who should have access to control such a system, and what measures could be implemented to prevent unauthorized access?
    - Could automation lead to over-reliance on technology? What are the risks and benefits?